IV IMMUNOGLOBULIN (IVIG)

Infusion rate titration

Thrombosis & renal risk

CHEMICAL NAME Pooled polyclonal human IgG (>95%) from thousands of healthy donors — 5% or 10% solution

CLASS Human normal immunoglobulin (passive immunisation; immunomodulator)

TRADE NAMES Privigen · Octagam · Gamunex · Kiovig · Hizentra (SC)

THERAPEUTIC USES / INDICATIONS

- Primary and secondary immunodeficiency (replacement therapy).
- Immune thrombocytopenia (ITP), Kawasaki disease (with aspirin), Guillain–Barré syndrome, CIDP, myasthenic crisis.
- Post-exposure prophylaxis: measles, hepatitis A, varicella (where VZIG unavailable) — IM/IV.
- Toxic shock syndrome, MIS-C, autoimmune encephalitis (specialist).

ADMINISTRATION

- **IV** infusion — start slowly (e.g. 0.5–1 mL/kg/h) and increase every 15–30 min if tolerated to max rate per product.
- Pre-hydrate (renal/thrombosis risk); premedication (paracetamol, antihistamine) if previous reactions.
- Dedicated line; do not shake; bring to room temperature.
- **SC** formulations for home replacement therapy. Dose by ideal/adjusted body weight in obesity.

DOSAGES

- **Immunodeficiency replacement:** 0.4–0.6 g/kg every 3–4 weeks (trough IgG target).
- **ITP:** 1 g/kg for 1–2 days (or 0.4 g/kg for 5 days).
- **Kawasaki disease:** 2 g/kg single infusion over 10–12 h.
- **Guillain–Barré:** 0.4 g/kg/day for 5 days (total 2 g/kg).

PHARMACOKINETICS

- **Peak:** end of infusion. Distributes intravascularly then extravascularly.
- **t½:** ~3–4 weeks (shorter in protein-losing states).

PHARMACODYNAMICS

- Replaces **antibodies** (immunodeficiency).
- Immunomodulation: blocks Fc receptors on macrophages (↓ platelet destruction), neutralises autoantibodies, inhibits complement, modulates cytokines.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Infusion reactions: headache, chills, fever, flushing, myalgia, nausea (rate-related).
- Fatigue, back pain.
- Post-infusion headache.

ADVERSE EFFECTS (SERIOUS)

- Thrombosis (MI, stroke, DVT); elderly, immobility, hyperviscosity.
- Acute kidney injury (sucrose-containing products, pre-existing renal disease, dehydration).
- Aseptic meningitis; haemolysis (high doses, non-O blood groups).
- Anaphylaxis (IgA deficiency with anti-IgA); TRALI; fluid overload.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to human immunoglobulin.
- Selective IgA deficiency with anti-IgA antibodies (use IgA-depleted product).
- **Caution:** thrombosis risk, renal impairment, heart failure, elderly.

KEY INTERACTIONS

- Live vaccines (MMR, varicella, yellow fever): reduced response; delay for 3–11 months depending on dose.
- **Nephrotoxic drugs:** ↑ AKI risk.

NURSING MANAGEMENT

- Baseline vital signs, weight, renal function, FBC; hydration.
- Vital signs before, 15 min after start, with each rate increase, and at end; slow/stop for reactions.
- Monitor urine output, creatinine, signs of thrombosis (chest pain, limb swelling, neuro signs), haemolysis (Hb).
- Record product name and batch (traceability); do not switch brands without reason.

PATIENT EDUCATION

- Report headache, chest pain, breathlessness, leg swelling, dark urine, stiff neck or rash.
- Drink plenty of fluids before and after the infusion.
- Live vaccines may need to be delayed for several months.

KEY POINTS

- Pooled human IgG — replacement and immunomodulation.
- Start slowly; titrate rate.
- ITP, Kawasaki (2 g/kg), GBS, immunodeficiency.
- Thrombosis, AKI, aseptic meningitis; batch traceability.